

Hierarchical backstepping-based control of a Gun Launched MAV in crosswinds: Theory and experiment[☆]



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ABSTRACT

This paper considers the problem of controlling the position and the orientation of a Gun Launched Micro Aerial Vehicle – GLMAV – despite unknown aerodynamic efforts. The proposed approach overcomes the problem of gyroscopic coupling by taking advantage from the structure of the thrust mechanism, which is made up of two counter rotating propellers. An adaptive hierarchical controller is designed, allowing the trajectory tracking and the stabilization of the vehicle's position and orientation while the unknown aerodynamic efforts are estimated by means of an identifier. The overall process is shown to be stable for constant, or slowly time varying, aerodynamic efforts. However numerical simulations demonstrate the satisfying controller's performance even with nonconstant aerodynamic efforts. Experimental results are also provided.

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1. Introduction

The requirements of autonomous navigation strategies for unmanned Micro Aerial Vehicles (MAV) has now become very demanding, motivated by the recent technological advances in actuator miniaturization and embedded electronics (Castillo, Lozano, & Dzul, 2005; Lozano, 2010; Valavanis, 2007). Indeed, they are expected to perform hovering flights as well as aggressive flights despite possible wind perturbations. Throughout the existing mechanical configurations described in the literature (Ganguli, 2004; Nonami, Kendoul, Suzuki, Wang, & Nakazawa, 2010), rotary-wing MAV is currently the most promising solution to this dual requirement. Among them, the architecture with two coaxial contra-rotating rotors was chosen because it best meets the needs of the GLMAV project.

The GLMAV concept consists in bringing a MAV very quickly onto the site where it begins to be operational by using the energy delivered by an external device. Thereby, the embedded energy is conserved for the autonomous flight only. The principle, illustrated in Fig. 1, is divided into three phases of flight. In the first one, the GLMAV, packaged into the projectile shell, is launched by a portable weapon and follows a ballistic trajectory to its apogee. In the second phase of flight, the projectile is transformed into a MAV through the deployment of the rotors. Finally, the GLMAV fulfills its observation mission in the autonomous flight mode.

Previous studies (Gnemmi & Haertig, 2008; Gnemmi, Koehl, Martinez, Changey, & Theodoulis, 2009; Koehl, Rafaralahy, Boutayeb, & Martinez, 2010; Wereley & Pines, 2001) have demonstrated the GLMAV concept feasibility based on theoretical and experimental investigations concerning the sizing of the device, the payload and autonomy, the energy necessary to launch the projectile containing the MAV and the dimensioning of the portable weapon. These studies also justified the choice of the two bladed coaxial contrarotating rotor aeromechanical configuration with cyclic swashplate, which offers both advantages and disadvantages. More information about coaxial rotor aeromechanical analysis could be found in the literature (Hall, Wong, & Auld, 2008; Ramasamy, Lee, & Leishman, 1994).

In the literature, both linear and nonlinear approaches have already been undertaken to control UMAV. The classic linear designs such as PID controllers (Kim & Shim, 2003), PD controllers (Mettler, 2003) and cascaded PID controllers (Lipera et al., 2001) disregard the input coupling and are effective application-oriented methods. The $\mathcal{H}_\infty$ control scheme (McFarlane & Glover, 1992), effective to cope with both model uncertainty and disturbance rejection, has also been successfully tested on a wide range of UMAVs (Civita, Papageorgiou, Messner, & Kanade, 2006; Gadewadikar, Lewis, Subbarao, & Chen, 2008). An interesting comparative study between several linear designs is given in Weilenmann, Christen, and Geering (1994). But in those works, based on local linearization, the proof of convergence is not guaranteed when the vehicle leaves its flight domain and the control approaches have limited capability to alleviate the coupling among variables.

More recently, interest has been focused on nonlinear globally stable control laws. However, designing detailed models of UMAV

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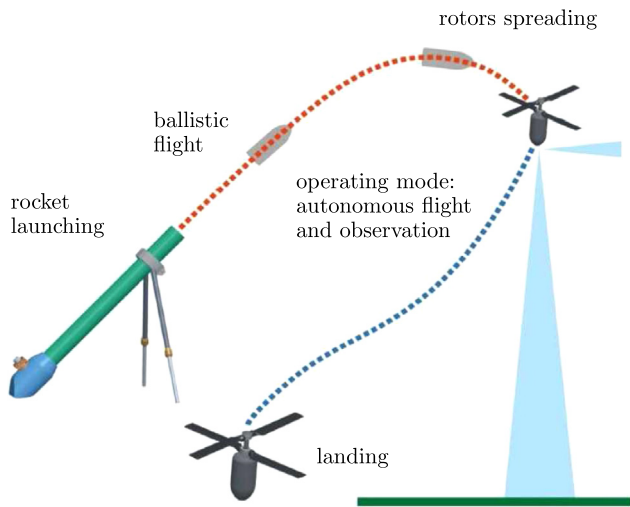


Fig. 1. The concept of the GLMAV.

nonlinear dynamics is unsuitable for control design purposes (Prouty, 1995). Therefore, a simplified nonlinear model of the dynamics was introduced in Koo and Sastry (1998), ignoring the input coupling. The resulting approximate model can be controlled via high gain designs, leading to a time scale separation between the cascaded systems (Mahony, Hamel, & Dzul, 1999), or backstepping designs, using dynamic extension of the thrust input (Mahony & Hamel, 2004), and dynamic inversion designs, leading to input–output linearization (Koo & Sastry, 1998). One of the difficulties in those designs is to deal with uncertain aerodynamic effects which may act on the vehicle. It has been overcome in Lipera et al. (2001), where the unwanted aerodynamic effects have been tabulated through intense wind tunnel tests, or in Isidori, Marconi, and Serrani (2003), which guarantees bounded tracking in the presence of parametric and model uncertainties.

This paper focuses on an adaptive controller, composed of a hierarchical control algorithm connected to an identification structure. This approach involves coming up with a high level position control and a low level attitude control. This architecture of controller is usual in most of the guidance, navigation and control algorithms (Isidori et al., 2003; Marconi & Naldi, 2007; Shim et al., 2000). That design is based on backstepping strategies (Krstic, Kanellakopoulos, & Kokotovic, 1995), in which control Lyapunov functions are derived, allowing the stabilization of the vehicle's position and orientation. The question of modeling the aerodynamic effects is overthrown by considering them as unknown perturbations that act on the system. In Drouot, Richard, and Boutayeb (2013) and Pflimlin, Soueres, and Hamel (2007), it was shown that an appropriate choice of adaptive filters in a backstepping process allows the estimation of unknown constant aerodynamic efforts. The main contribution of the paper is to introduce a control structure that handles unknown variable aerodynamic efforts.

The paper is organized as follows: the architecture of the vehicle is briefly introduced in Section 2, the nonlinear model of the GLMAV is developed in Section 3 and the control design is described in Section 4. Sections 5 and 6 are devoted to present the results of the numerical simulations and the experiments respectively. Finally, the last section contains concluding remarks and a discussion on further works.

2. The Gun Launched MAV

From a mechanical point of view, the preliminary version of the GLMAV is a 350 mm in wing span, 400 mm in height coaxial

helicopter with a mass of approximately 300 g. The diameter of the body of the GLMAV is in turn of 80 mm. The two rotors are driven by two brushless DC motors and they rotate in opposite ways to overcome the gyroscopic effect. This particular architecture allows to control the yaw angle by a differential speed variation between the two rotors, whereas the altitude which is controlled by simultaneously varying the rotor speeds. The vehicle is steered by a conventional swashplate placed on the lower rotor, that is actuated by two servomotors. All the signals to the motors and servomotors controllers are pulse width modulated signals.

The sensors embedded on the platform are part of a SBG Systems IG-500N Inertial Measurement Unit completed with a barometer and a 4 Hz GPS receiver. With this setup, the position, the velocity, the attitude and the angular velocity of the vehicle can theoretically be reliably measured at a rate of 100 Hz. The data fusion algorithm used in that IMU is an Extended Kalman Filter whose features are not accessible. Further, the GLMAV is equipped with a down- and side-looking camera which might eventually be used for position or velocity measurement. The electronic architecture relies on a Gumstix Overo Fire single-board computer working with an OMAP3530 processor. That computer is equipped with a Linux operating system and with a DSP executing the digital signal processing tasks (computation of the control, filtering, etc.) at a high rate. Regarding the communication, the ZigBee protocol is used to send the high level instructions from the remote and the WiFi protocol is used to relay information as the data from the sensors, the speed of the rotors, the battery status, etc. Finally, the components of the GLMAV are powered by a lithium polymer battery located at the bottom of the vehicle (see Fig. 2).

3. Nonlinear model

The nonlinear mathematical model developed for the GLMAV is based on the physical laws of mechanics and aeromechanics. Indeed, early studies (Gnemmi et al., 2009; Krashanitsa, Platanitis, Silin, & Shkarayev, 2006) have shown that MAV models can be divided into two submodels, i.e. a six degrees of freedom mechanical model coupled with an aerodynamic model. The objective of the model is to be as simple as possible, since it has to be used for the control design. On the other hand, the physics and dynamics of

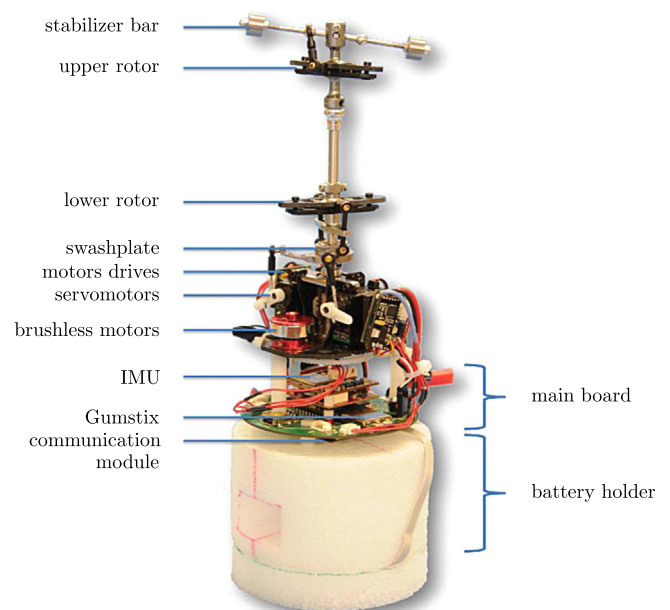


Fig. 2. The first prototype of the GLMAV.

the different components of the GLMAV have to be reflected accurately.

3.1. Mechanical model

As common in aeronautics, an inertial frame $\mathcal{I}$ and a body fixed frame $\mathcal{B}$ are introduced. Using the Newtonian mechanics, the differential equations of the dynamics of the vehicle with respect to the inertial frame is described by

$$\begin{aligned}\dot{\xi} &= v \\ m\dot{v} &= R_\eta f \\ \dot{\eta} &= Q_\eta \Omega \\ J\dot{\Omega} &= -\Omega_\times J\Omega + \Gamma\end{aligned}\quad (1)$$

where ξ and v represent respectively the position and the velocity of the vehicle, m its mass which is assumed constant, J its diagonal inertia matrix, Ω its angular velocity expressed in the body frame, f and Γ the forces and torques applied to the vehicle also expressed in the body frame and R_η the rotation matrix relating the two reference frames. The matrices R_η and Q_η can be parameterized by means of the Euler angles η obtaining

$$R_\eta = \begin{bmatrix} c_\theta c_\psi & s_\phi s_\theta c_\psi - c_\phi s_\psi & c_\phi s_\theta c_\psi + s_\phi s_\psi \\ c_\theta s_\psi & s_\phi s_\theta s_\psi + c_\phi c_\psi & c_\phi s_\theta s_\psi - s_\phi c_\psi \\ -s_\theta & s_\phi c_\theta & c_\phi c_\theta \end{bmatrix}\quad (2)$$

and

$$Q_\eta = \begin{bmatrix} 1 & s_\phi t_\theta & c_\phi t_\theta \\ 0 & c_\phi & -s_\phi \\ 0 & s_\phi/c_\theta & c_\phi/c_\theta \end{bmatrix}.\quad (3)$$

For convenience, it has been noted $c_\varrho = \cos \varrho$, $s_\varrho = \sin \varrho$, $t_\varrho = \tan \varrho$ and $\Omega_\times$ stands for the skew-symmetric related to Ω .

So far, the equations of motion are independent from the aerodynamic structure and can be easily found in the literature. In order to complete that model, the force f and the moment Γ vectors have to be defined.

3.2. Aerodynamic model

The aerodynamic model focuses on the forces and torques generated by the coaxial rotors and the cyclic swashplate incidence angles. The gyroscopic moments induced by the contrarotating rotors are neglected because they offset each other assuming that the speed differential is virtually zero. Further explanations on the aerodynamic modeling can be found in Koehl et al. (2010).

3.2.1. Forces acting on the GLMAV

The main force that acts on the vehicle is the thrust generated by the two rotors. The upper rotor contributes only to the vertical thrust whereas the lower rotor, which generates both vertical and lateral forces due to the swashplate incidence angles. The total force T generated by the coaxial rotors is in $\mathcal{B}$

$$T = \begin{bmatrix} T_x \\ T_y \\ T_z \end{bmatrix} = \begin{bmatrix} -\beta \sin \delta_{c_y} \cos \delta_{c_x} \Omega_2^2 \\ -\beta \sin \delta_{c_x} \Omega_2^2 \\ \alpha \Omega_1^2 + \beta \cos \delta_{c_x} \cos \delta_{c_y} \Omega_2^2 \end{bmatrix}\quad (4)$$

where α and β , with $\alpha < 0$ and $\beta < 0$, are the rotors aerodynamic coefficients, Ω_1 and Ω_2 the rotors rotation speeds and δ_{c_x} and δ_{c_y} the swashplate incidence angles.

In addition to the thrust, the weight and some aerodynamic forces also act on the vehicle. As usual in the literature, the weight

force is written in $\mathcal{B}$ as

$$f_w = mg R_\eta^T e_3. \quad (5)$$

The aerodynamic forces, mainly due to the different drags, are difficult to determine because of their dependence on the wind, which is unpredictable. This is why the net force of these efforts will be denoted by f_{ext} in $\mathcal{B}$ and by F_{ext} in $\mathcal{I}$.

Adding the different forces that act on the GLMAV leads to

$$f = [X \ Y \ Z]^T = T + f_w + f_{ext}. \quad (6)$$

3.2.2. Moments acting on the GLMAV

A torque on the longitudinal or lateral axis may be produced via differentially pitching the swashplate incidence angles δ_{c_x} and δ_{c_y} . On the other hand, the torque on the vertical axis is induced by the rotation speed difference between the two rotors. The roll, pitch and yaw moments are then given by

$$\tau = \begin{bmatrix} \tau_L \\ \tau_M \\ \tau_N \end{bmatrix} = \begin{bmatrix} -d\beta \sin \delta_{c_x} \Omega_2^2 \\ d\beta \sin \delta_{c_y} \cos \delta_{c_x} \Omega_2^2 \\ \gamma_1 \Omega_1^2 + \gamma_2 \Omega_2^2 \end{bmatrix}\quad (7)$$

where d is the distance between the center of gravity and the lower rotor center of rotation and γ_1 and γ_2 , with $\gamma_1 > 0$ and $\gamma_2 < 0$, are the yaw aerodynamic coefficients.

In addition, the aerodynamic forces induce nonnegligible moments that apply at a point difficult to determine. However, because of the symmetrical property of the GLMAV, it is assumed that this point is located somewhere on the z_B axis, but shifted away from the center of gravity. The moments will be denoted by M_{ext} in the body frame. The weight force f_w , because it applies at the center of gravity, does not induce any moment.

Adding the different moments that act on the GLMAV leads to

$$\Gamma = [L \ M \ N]^T = \tau + M_{ext}. \quad (8)$$

The different variables related to the model of the vehicle are gathered in Fig. 3.

3.3. Approximate nonlinear model

Combining (6) and (8) with (1) leads to the nonlinear model of the dynamics of the GLMAV that will be used in the future

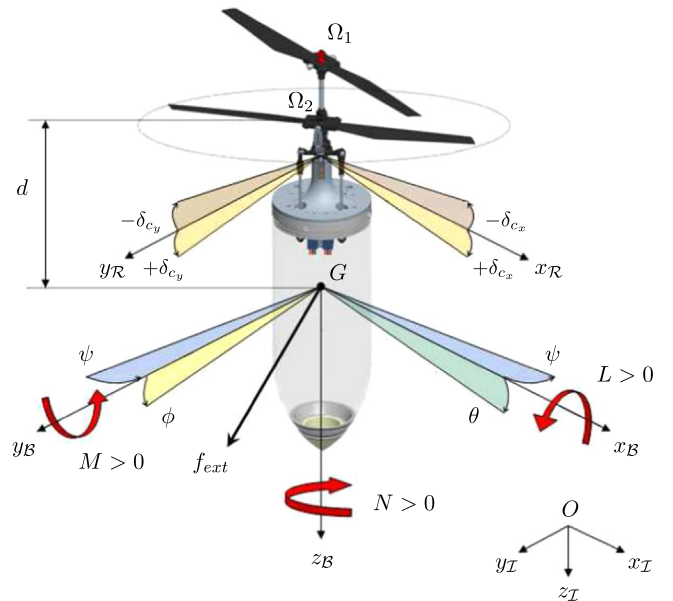


Fig. 3. The variables of the GLMAV.

numerical simulations. It is written as

$$\begin{aligned}\dot{\xi} &= v \\ m\dot{v} &= R_\eta T + mge_3 + F_{ext} \\ \dot{\eta} &= Q_\eta \Omega \\ J\dot{\Omega} &= -\Omega \times J\Omega + \tau + M_{ext}\end{aligned}\quad (9)$$

However, in order to simplify the theoretical analysis and hence the synthesis of the control law, a certain number of approximations shall be considered. In particular the lateral forces T_x and T_y in (4) will be assumed negligible compared to the contribution of main thrust T_z . Therefore, the simulation model (9) becomes

$$\begin{aligned}\dot{\xi} &= v \\ m\dot{v} &= T_z R_\eta e_3 + mge_3 + F_{ext} \\ \dot{\eta} &= Q_\eta \Omega \\ J\dot{\Omega} &= -\Omega \times J\Omega + \tau + M_{ext}\end{aligned}\quad (10)$$

Nevertheless, this approximate model still has 12 states and is still governed by the control inputs Ω_1 and Ω_2 , the upper and lower rotor rotational speeds, and δ_{cx} and δ_{cy} , the cyclic swashplate incidence angle along the roll and pitch axes. Using the third component of (4) and (7), those control inputs are given by

$$\begin{aligned}\Omega_1^2 &= (\gamma_2 T_z - \beta \tau_n) / (\alpha \gamma_2 - \beta \gamma_1) \\ \Omega_2^2 &= (\alpha \tau_n - \gamma_1 T_z) / (\alpha \gamma_2 - \beta \gamma_1) \\ \delta_{cx} &= -\tau_L / d\beta \Omega_2^2 \\ \delta_{cy} &= \tau_M / d\beta \Omega_2^2\end{aligned}\quad (11)$$

with T_z , τ_L , τ_M , τ_n the new inputs to be determined. The computations were done assuming that the swashplate incidence angles were small enough, meaning that the paraxial approximation was used.

4. Control design

The design of the nonlinear hierarchical control law that will be applied on the GLMAV is based on backstepping techniques (Krstic et al., 1995). Its aim is to shape the new inputs T_z , τ_L , τ_M , τ_n in order to track the desired longitudinal, lateral, vertical and heading time references $x_d(t)$, $y_d(t)$, $z_d(t)$ and $\psi_d(t)$. The reference signals are supposed to be known arbitrary time profiles with the only restrictions dictated by the functional controllability of the system, by the fulfillment of the flight envelope constraints and of physical constraints on the control inputs.

A hierarchical control approach has now become quite common for controlling UAVs, whose orientation dynamics is faster than the translation dynamics. Indeed, in this control structure, the lowest level, with the highest bandwidth, is in control of the vehicle orientation and the top level is in control of the vehicle position along a given trajectory. These two levels form nested feedback loops, as shown in Fig. 4, with the output of the outer loop being the input of the inner loop. An interesting aspect of that specific architecture is that if a disturbance affects the inner loop, it will be handled by the inner controller before it affects the outer loop, hence offering great robustness properties. Therefore, this simple control structure enables simple, but very effective control algorithms. Furthermore, when using the backstepping design approach, it will be shown that the hierarchical architecture reduces the complexity of the control signals.

The choice of backstepping techniques over other nonlinear control design methods is also quasi-self-evident. Indeed, an appealing feature of these techniques is that they do not force the designed system to appear linear, avoiding cancellations of useful nonlinearities. Another interesting point when using backstepping designs is that the stability analysis of the closed-loop

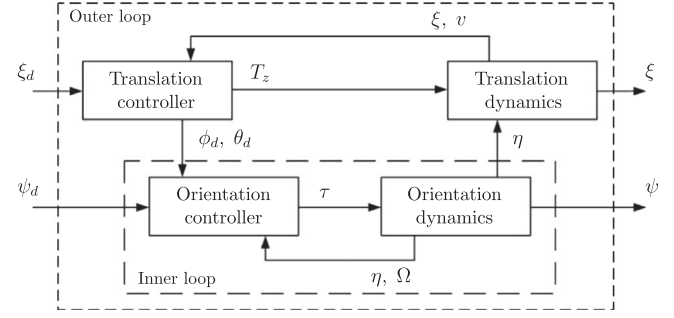


Fig. 4. Structure of the hierarchical control law.

system, such as the determination of the type of convergence, are simple thanks to the Lyapunov stability theory. Directly compared to some other nonlinear control design methods, the backstepping approach has many advantages. Indeed, compared to nested saturations controllers (Isidori et al., 2003), there are fewer control parameters to be used and their value is much easily chosen. Moreover, compared to sliding mode controllers (McGeoch, McGookin, & Houston, 2005), the backstepping approach is far less sensitive to actuators flaws. Finally, compared to controllers based on nonlinear $\mathcal{H}_\infty$ designs (Kung, Yang, Chiou, & Luo, 2002) or predictive techniques (Grune & Pannek, 2011), the computational complexity is greatly reduced. For all these reasons, the backstepping techniques seem to be the most appropriate nonlinear control approach for flying an UAV.

In this study, it will be assumed that all the states are accessible for control purpose, in particular vectors ξ and v for the translation dynamics, and vectors η and Ω for the orientation dynamics. Furthermore the initial state is supposed to belong to any (arbitrarily small) compact set with the only restriction that the paraxial approximation can be used.

4.1. Control structure

The hierarchical backstepping control design procedure is now developed, leaving the identification issues for the next section. Introducing the first error variable as

$$\delta_1 = \xi - \xi_d, \quad (12)$$

corresponding to the position tracking error, the first control Lyapunov function is defined by

$$V_1 = \frac{1}{2} \delta_1^T \delta_1, \quad (13)$$

with the time derivative given by

$$\dot{V}_1 = \delta_1^T (v - \dot{\xi}_d), \quad (14)$$

here v appears as the first virtual input and its desired value is chosen to be

$$v_d = -k_{\delta_1} \delta_1 + \dot{\xi}_d, \quad (15)$$

along with the corresponding error variable

$$\delta_2 = v - v_d \quad (16)$$

such that (14) may be written as

$$\dot{V}_1 = -k_{\delta_1} \|\delta_1\|^2 + \delta_1^T \delta_2. \quad (17)$$

Backstepping an integrator, the second control Lyapunov function is defined by

$$V_2 = V_1 + \frac{1}{2} \delta_2^T \delta_2. \quad (18)$$

Differentiating (18) with respect to time leads to

$$\dot{V}_2 = -k_{\delta_1} \|\delta_1\|^2 + \delta_1^T \delta_2 + \delta_2^T \left(\frac{1}{m} T_z R_{\eta} e_3 + g e_3 + \frac{1}{m} \hat{F}_{ext} - \dot{v}_d \right) + \frac{1}{m} \delta_2^T \tilde{F}_{ext} \quad (19)$$

where e_3 is the third vector of the canonical basis $\mathbb{R}^3$ and $\tilde{F}_{ext}$ is the estimation error between F_{ext} and its estimate $\hat{F}_{ext}$.

Here, η and T_z appear as the second virtual inputs and their desired values are chosen to be

$$\frac{1}{m} T_{z_d} R_{\eta_d} e_3 + g e_3 + \frac{1}{m} \hat{F}_{ext} = -\delta_1 - k_{\delta_2} \delta_2 + \dot{v}_d = a_d \quad (20)$$

where $\eta_d = [\phi_d \ \theta_d \ \psi_d]^T$ and $a_d = [a_{d_x} \ a_{d_y} \ a_{d_z}]^T$ can be interpreted as a virtual, desired acceleration. Its explicit expression is given in (30).

The definition of the next error variable follows as

$$\delta_3 = \frac{1}{m} T_z R_{\eta} e_3 - \frac{1}{m} T_{z_d} R_{\eta_d} e_3. \quad (21)$$

Combining (20) and (21) with (19) leads to

$$\dot{V}_2 = -k_{\delta_1} \|\delta_1\|^2 - k_{\delta_2} \|\delta_2\|^2 + \delta_2^T \delta_3 + \frac{1}{m} \delta_2^T \tilde{F}_{ext}. \quad (22)$$

The conventional backstepping method (Krstic et al., 1995) would require the definition of a third control Lyapunov function by augmenting V_2 with a positive definite function of δ_3 . But this would require taking a dynamic extension of the control input T_z (Isidori, 1989). In other words, the system input T_z and its derivative $\dot{T}_z$ would then be viewed as internal states of the controller. This would also lead to more complex control equations than the ones that are presented in this paper. Instead it is explicitly set

$$T_z = T_{z_d}. \quad (23)$$

From (20), it is straightforward to show that

$$T_{z_d} = -m \sqrt{\tilde{a}_{d_x}^2 + \tilde{a}_{d_y}^2 + (\tilde{a}_{d_z} - g)^2}, \quad (24)$$

$$\begin{cases} \phi_d = \arctan \left(\frac{\tilde{a}_{d_y}}{\sqrt{\tilde{a}_{d_x}^2 + (\tilde{a}_{d_z} - g)^2}} \right) \\ \theta_d = \arctan \left(\frac{\tilde{a}_{d_x}}{\tilde{a}_{d_z} - g} \right) \end{cases} \quad \text{for } \psi_d \approx 0 [2\pi], \quad (25)$$

$$\begin{cases} \phi_d = -\arctan \left(\frac{\tilde{a}_{d_x}}{\sqrt{\tilde{a}_{d_y}^2 + (\tilde{a}_{d_z} - g)^2}} \right) \\ \theta_d = \arctan \left(\frac{\tilde{a}_{d_y}}{\tilde{a}_{d_z} - g} \right) \end{cases} \quad \text{for } \psi_d \approx \frac{\pi}{2} [2\pi], \quad (26)$$

$$\begin{cases} \phi_d = -\arctan \left(\frac{\tilde{a}_{d_y}}{\sqrt{\tilde{a}_{d_x}^2 + (\tilde{a}_{d_z} - g)^2}} \right) \\ \theta_d = -\arctan \left(\frac{\tilde{a}_{d_x}}{\tilde{a}_{d_z} - g} \right) \end{cases} \quad \text{for } \psi_d \approx \pi [2\pi] \quad (27)$$

and

$$\begin{cases} \phi_d = \arctan \left(\frac{\tilde{a}_{d_x}}{\sqrt{\tilde{a}_{d_y}^2 + (\tilde{a}_{d_z} - g)^2}} \right) \\ \theta_d = -\arctan \left(\frac{\tilde{a}_{d_y}}{\tilde{a}_{d_z} - g} \right) \end{cases} \quad \text{for } \psi_d \approx -\frac{\pi}{2} [2\pi] \quad (28)$$

with

$$\tilde{a}_d = a_d - \frac{1}{m} \hat{F}_{ext} \quad (29)$$

where $T_{z_d} \in \mathbb{R}$, $\phi_d \in [-\pi/2, \pi/2]$ and $\theta_d \in [-\pi/2, \pi/2]$ are the solutions to (20). Two possible solutions exist for these virtual controls. For obvious practical reasons, we chose the one corresponding to negative primary thrust (i.e. $T_{z_d} < 0$). The virtual, desired acceleration a_d is determined from (20), by replacing δ_1 and δ_2 by their expression respectively given by (12) and (16):

$$a_d = -(k_{\delta_1} k_{\delta_2} + 1)(\xi - \xi_d) - (k_{\delta_1} + k_{\delta_2})v + (k_{\delta_1} + k_{\delta_2})\dot{\xi}_d + \ddot{\xi}_d. \quad (30)$$

A new error variable is then defined as

$$\varepsilon_1 = \eta - \eta_d \quad (31)$$

and the backstepping method is carried on by designing a controller that will force ε_1 to converge to zero. Because of the change in error variables, it will not be possible to cancel the cross term $\delta_2^T \delta_3$ in (22). However, from (21) and (23), we are assured that as $\varepsilon_1 \rightarrow 0$, $\delta_3 \rightarrow 0$.

Proceeding with the control design, the third control Lyapunov function is defined by

$$V_3 = \frac{1}{2} \varepsilon_1^T \varepsilon_1 \quad (32)$$

with the time derivative given by

$$\dot{V}_3 = \varepsilon_1^T (Q_{\eta} \Omega - \dot{\eta}_d). \quad (33)$$

Here Ω appears as the third virtual input and its desired value is chosen to be

$$Q_{\eta} \Omega_d = -k_{\varepsilon_1} \varepsilon_1 + \dot{\eta}_d. \quad (34)$$

The definition of the fourth error variable follows as

$$\varepsilon_2 = Q_{\eta} (\Omega - \Omega_d) \quad (35)$$

such that

$$\dot{V}_3 = -k_{\varepsilon_1} \|\varepsilon_1\|^2 + \varepsilon_1^T \varepsilon_2. \quad (36)$$

Backstepping the final integrator, the last control Lyapunov function is defined as

$$V_4 = V_3 + \frac{1}{2} \varepsilon_2^T \varepsilon_2. \quad (37)$$

Differentiating (37) with respect to time leads to

$$\dot{V}_4 = -k_{\varepsilon_1} \|\varepsilon_1\|^2 + \varepsilon_1^T \varepsilon_2 + \varepsilon_2^T (\dot{Q}_{\eta} (\Omega - \Omega_d) + Q_{\eta} (\dot{\Omega} - \dot{\Omega}_d)). \quad (38)$$

According to (1), (38) can be rewritten as

$$\dot{V}_4 = -k_{\varepsilon_1} \|\varepsilon_1\|^2 + \varepsilon_1^T \varepsilon_2 + \varepsilon_2^T (\dot{Q}_{\eta} (\Omega - \Omega_d) - Q_{\eta} \dot{\Omega}_d) + \varepsilon_2^T Q_{\eta} J^{-1} (-\Omega \times J \Omega + \tau + \hat{M}_{ext}) + \varepsilon_2^T Q_{\eta} J^{-1} \tilde{M}_{ext}, \quad (39)$$

where $\tilde{M}_{ext}$ is the estimation error between M_{ext} and its estimate $\hat{M}_{ext}$.

As the physical control input τ has appeared, it is defined as

$$\tau = \Omega \times J \Omega - \hat{M}_{ext} + J Q_{\eta}^{-1} (Q_{\eta} \dot{\Omega}_d - \dot{Q}_{\eta} (\Omega - \Omega_d) - \varepsilon_1 - k_{\varepsilon_2} \varepsilon_2) \quad (40)$$

such that

$$\dot{V}_4 = -k_{\varepsilon_1} \|\varepsilon_1\|^2 - k_{\varepsilon_2} \|\varepsilon_2\|^2 + \varepsilon_2^T Q_{\eta} J^{-1} \tilde{M}_{ext}. \quad (41)$$

One can therefore easily notice from (29) and (40) that an explicit expression is required for $\hat{F}_{ext}$ and $\hat{M}_{ext}$. Furthermore, the error terms $\tilde{F}_{ext}$ and $\tilde{M}_{ext}$ in (22) and (41) have to be cancelled. As these elements are unknown due to their dependence on the wind, a specific identifier has to be used.

4.2. Identification structure

Consider the system

$$\dot{x} = f(x, \zeta) + g(x, \zeta)u \quad (42)$$

where $x \in \mathbb{R}^n$ denotes the state of the system, $u \in \mathbb{R}$ is its control input and $\zeta \in \mathbb{R}^p$ is a vector of unknown but constant parameters. The state function f and the input function g are assumed to be smooth vector fields on $\mathbb{R}^n$. Further, let $f(x, \zeta)$ and $g(x, \zeta)$ have the

form

$$f(x, \zeta) = f_0(x) + \sum_{i=1}^p f_i(x) \zeta_i \quad (43)$$

$$g(x, \zeta) = g_0(x) + \sum_{i=1}^p g_i(x) \zeta_i. \quad (44)$$

Here ζ_i , $i = 1, \dots, p$, are the unknown parameters which appear linearly and the smooth vector fields $f_0(x)$, $f_i(x)$, $g_0(x)$ and $g_i(x)$ are known. Let the regressor be

$$w^T(x, u) = [f_1(x) + g_1(x)u, \dots, f_p(x) + g_p(x)u], \quad (45)$$

then (42) can be written as

$$\dot{x} = w_0(x, u) + w^T(x, u)\zeta \quad (46)$$

where

$$w_0(x, u) = f_0(x) + g_0(x)u. \quad (47)$$

To estimate the unknown parameters, the following identifier system is used:

$$\begin{aligned} \dot{\hat{x}} &= A(\hat{x} - x) + w_0(x, u) + w^T \Lambda^{-1}(x, u) \hat{\zeta} \\ \dot{\hat{\zeta}} &= -w(x, u)P(\hat{x} - x) - \chi(x), \end{aligned} \quad (48)$$

where $\hat{x}$ is the observer state, $\hat{\zeta}$ is the estimated parameter, $\chi(x)$ is a vanishing term relating the identification structure to the hierarchical controller, $A = \text{diag}(A_1, A_2) \in \mathbb{R}^{n \times n}$ is a Hurwitz matrix with a set of eigenvalues with real parts less than $-\sigma$ ($\sigma > 0$), $\Lambda = \Lambda^T = \text{diag}(\Lambda_1, \Lambda_2) > 0$ is a $(n/p) \times (n/p)$ diagonal matrix which tunes up the adaptation dynamics of $\hat{x}$ and $P = P^T = \text{diag}(P_1, P_2) \in \mathbb{R}^{n \times n} > 0$ is a solution to the Lyapunov equation:

$$A^T P + P A = -Q, \quad Q = \text{diag}(Q_1, Q_2) > 0. \quad (49)$$

This identifier structure is reminiscent of one proposed in Kudva and Narendra (1973) and Sheikholeslam (1995). Defining $\tilde{x} = \hat{x} - x$ as the observer state error, and $\tilde{\zeta} = \hat{\zeta} - \zeta$ as the parameter error, and assuming ζ to be constant but unknown, then the following error system can be stated:

$$\begin{aligned} \dot{\tilde{x}} &= A\tilde{x} + w^T(x, u)\Lambda^{-1}\tilde{\zeta} \\ \dot{\tilde{\zeta}} &= -w(x, u)P\tilde{x} - \chi(x) \end{aligned} \quad (50)$$

The proof of stability of the observer-based identifier alone, i.e. when $\chi(x) = 0$, is a standard Lyapunov argument on the function:

$$V(\tilde{x}, \tilde{\zeta}) = \tilde{x}^T P \tilde{x} + \tilde{\zeta}^T \Lambda^{-1} \tilde{\zeta} \quad (51)$$

which is decreasing along the trajectories of (50); thereby establishing bounded $\tilde{x}$ and $\tilde{\zeta}$. However, to prove that $\tilde{x} \rightarrow 0$ as $t \rightarrow \infty$, it must be shown that $\tilde{x}$ is uniformly continuous or, alternatively, that $\dot{\tilde{x}}$ is bounded. This in turn needs w to be bounded as well. And it was shown in Sastry and Bodson (1989) that if the system is bounded-input bounded-state stable with bounded input, then w is bounded. Furthermore, as it is standard in the literature (Dong & Mei, 2007; Kreisselmeier, 1977; Morgan & Narendra, 1977), if there exist constants ρ_1 , ρ_2 and δ such that

$$0 < \rho_1 I \leq \int_t^{t+\delta} w(s)w^T(s) ds \leq \rho_2 I \quad \text{for all } t \geq 0, \quad (52)$$

then the identifier system defined by (50) is globally exponentially stable, meaning that both $\tilde{x}$ and $\tilde{\zeta}$ converge to zero exponentially fast, with degree not less than $\min(\sigma, \mathcal{K})$, where

$$\mathcal{K} = \frac{\rho_1 \lambda_{\min}(P)}{(1 + n\rho_2 \lambda_{\max}(P))^2}, \quad (53)$$

with $\lambda_{\min}(P)$ and $\lambda_{\max}(P)$ standing respectively for the lowest and the highest eigenvalue of the matrix P . The expression (52) is known as a

persistently exciting condition, with I standing for the identity matrix of appropriate dimensions.

The application of the identification technique to the system (1) is direct by taking

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \xi \\ \frac{v}{\eta} \\ \Omega \end{bmatrix} \quad \text{and} \quad \zeta = \begin{bmatrix} \zeta_1 \\ \zeta_2 \end{bmatrix} = \begin{bmatrix} F_{ext} \\ M_{ext} \end{bmatrix} \quad (54)$$

with

$$w_0(x, u) = \begin{bmatrix} v \\ \frac{1}{m} T_z R_\eta e_3 + g e_3 \\ Q_\eta \Omega \\ J^{-1}(-\Omega \times J \Omega + \tau) \end{bmatrix}, \quad (55)$$

$$w(x, u) = \begin{bmatrix} w_1(x, u) \\ w_2(x, u) \end{bmatrix} = \begin{bmatrix} 0_{3 \times 3} \frac{1}{m} I_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 3} \\ 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 3} J^{-1} \end{bmatrix}, \quad (56)$$

$$\chi(x) = \begin{bmatrix} \chi_1(x) \\ \chi_2(x) \end{bmatrix} = \begin{bmatrix} \frac{1}{2m} \Lambda_1 \delta_2 \\ \frac{1}{2} \Lambda_2 (e_2^T Q_\eta J^{-1})^T \end{bmatrix}, \quad (57)$$

leading to the closed-loop observer-based control structure shown in Fig. 5.

4.3. Closed-loop stability and closed-loop dynamics

The proof of stability of the observer-based hierarchical control is achieved by augmenting the Lyapunov functions (18) and (27) with a positive definite function of $\tilde{x}$ and $\tilde{\zeta}$. Indeed, considering

$$V_2 = V_1 + \frac{1}{2} \delta_2^T \delta_2 + \tilde{x}_1^T P_1 \tilde{x}_1 + \tilde{\zeta}_1^T \Lambda_1^{-1} \tilde{\zeta}_1 \quad (58)$$

and

$$V_4 = V_3 + \frac{1}{2} \varepsilon_2^T \varepsilon_2 + \tilde{x}_2^T P_2 \tilde{x}_2 + \tilde{\zeta}_2^T \Lambda_2^{-1} \tilde{\zeta}_2, \quad (59)$$

it is straightforward to show that their derivatives are given by

$$\dot{V}_2 = -k_{\delta_1} \|\delta_1\|^2 - k_{\delta_2} \|\delta_2\|^2 + \delta_2^T \delta_3 - \tilde{x}_1^T Q_1 \tilde{x}_1 \quad (60)$$

and

$$\dot{V}_4 = -k_{\varepsilon_1} \|\varepsilon_1\|^2 - k_{\varepsilon_2} \|\varepsilon_2\|^2 - \tilde{x}_2^T Q_2 \tilde{x}_2, \quad (61)$$

which are both negative semi definite functions. Furthermore, applying the control signals (24) and (40) to the system (1), the closed-loop translational dynamics may be written in error coordinates as

$$\begin{aligned} \dot{\delta}_1 &= -k_{\delta_1} \delta_1 + \delta_2 \\ \dot{\delta}_2 &= -\delta_1 - k_{\delta_2} \delta_2 + \delta_3 + \frac{1}{m} \tilde{F}_{ext} \end{aligned} \quad (62)$$

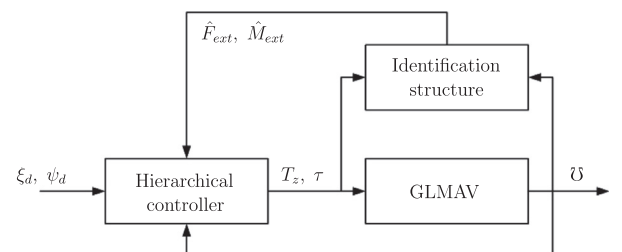


Fig. 5. Structure of the control process with $\tilde{f} = [\xi, v, \eta, \Omega]$.

As for the closed-loop orientation dynamics, it may be written in error coordinates as

$$\begin{aligned}\dot{\varepsilon}_1 &= -k_{\varepsilon_1} \varepsilon_1 + \varepsilon_2 \\ \dot{\varepsilon}_2 &= -\varepsilon_1 - k_{\varepsilon_2} \varepsilon_2 + Q_p J^{-1} \tilde{M}_{ext}\end{aligned}\quad (63)$$

Noting that δ_3 vanishes as $\varepsilon_1 \rightarrow 0$ (from (21) and (23)), the controlled system has the structure of an upper subsystem describing the translational dynamics (62) in cascade with a lower subsystem describing the orientation dynamics (63).

From the positive definite Lyapunov function (59) and its respective negative semi-definite derivative (61), it is clear that the error variables ε_1 and ε_2 , as well as their successive derivatives, asymptotically converge to zero. The application of LaSalle's invariance principle then guarantees the convergence of the estimation error $\tilde{M}_{ext}$ to the largest positively invariant set verifying $\dot{V}_4 = 0$. Looking at the expression of ε_2 given by (63), it is clear that $\tilde{M}_{ext}$ asymptotically converges to zero.

Furthermore, from the positive definite Lyapunov function (58) and its respective negative semi-definite derivative (60), it is clear that the error variables δ_1 and δ_2 , as well as their successive derivatives, asymptotically converge to zero. In addition, as previously mentioned, it is certain that the error variable δ_3 converges to zero as ε_1 converges to zero. The application of LaSalle's invariance principle then guarantees the convergence of the estimation error $\tilde{F}_{ext}$ to the largest positively invariant set verifying $\dot{V}_2 = 0$. Looking at the expression of δ_2 given by (62), it is clear that $\tilde{F}_{ext}$ asymptotically converges to zero.

4.4. Singularity issues

As it is designed, the controller contains a singularity which can be explicited by recalling the expression for the desired acceleration (20):

$$\frac{1}{m} T_{zd} R_{\eta_d} e_3 + g e_3 + \frac{1}{m} \hat{F}_{ext} = a_d. \quad (64)$$

A singularity will be encountered at every point along the closed-loop trajectory where the state configuration corresponding to $a_d(t) = g e_3 + (1/m) \hat{F}_{ext}$ occurs. It is obvious from (64) that at such instances $T_{zd} = 0$, and thus η_d becomes undefined. This singularity may cause significant problems. Indeed, the hierarchical backstepping-based controller forces the vehicle orientation η to converge to the desired η_d . It achieves this by feeding forward the η_d signal into the orientation dynamics using its first two derivatives. However, an inspection of the analytic expressions for $\dot{\eta}_d$ and $\ddot{\eta}_d$ reveals that as $a_d(t) \rightarrow g e_3 + (1/m) \hat{F}_{ext}$, $\dot{\eta}_d \rightarrow \infty$ and $\ddot{\eta}_d \rightarrow \infty$. Thus, closed-loop trajectories passing close to this singularity will demand exceptionally large values of orientation control signal τ . That is, it is possible for the desired orientation to go through a π radian discontinuity, resulting in unbound feedforward terms.

There exist some techniques mitigating this singularity. One, proposed in Olfati-Saber (2002), uses saturation within the controller such that the singularity never occurs. However, it is shown that this approach has the potential to limit the controller performance when rejecting large non-zero initial conditions (Wood, 2007). Another one, suggested in Wood (2007), modifies the controller structure whereby additional dynamics are embedded into the controller. These dynamics are designed such that the singularity never occurs.

5. Simulation results

Some numerical simulations were performed on the model (9) with physical parameters given in Table 1. Some of these parameters

Table 1
Parameters of the GLMAV.

Parameter	Value	Unit
m	0.410	kg
g	9.81	m s ⁻²
J_{xx}	1.383×10^{-3}	kg m ²
J_{yy}	1.383×10^{-3}	kg m ²
J_{zz}	2.72×10^{-4}	kg m ²
α	-3.6835×10^{-5}	N rad ⁻² s ⁻²
β	-3.7760×10^{-5}	N rad ⁻² s ⁻²
γ_1	1.4765×10^{-6}	N m rad ⁻² s ⁻²
γ_2	-1.3266×10^{-6}	N m rad ⁻² s ⁻²
d	0.0676	m

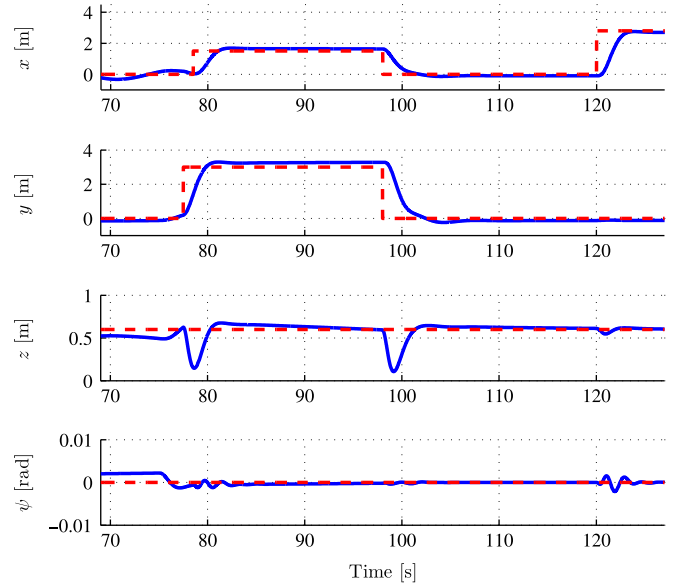


Fig. 6. Output signals x , y , z and ψ (solid blue lines) and reference signals (dashed red lines). (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)

were experimentally identified and validated in Koehl et al. (2010). The structure of the overall process tested in simulation is pictured in Fig. 5.

The simulation initial conditions are those expected at the beginning of the third phase of flight. Hence the velocity and the angular velocity, such as the roll and pitch angles, are set to 0. The initial position is chosen to be $\xi_0 = [0 \ 0 \ 0.6]^T$ m, while the yaw angle is initially set to $\psi_0 = 0$ rad. The control gains were selected such that the real part of the eigenvalues of the translation and orientation closed-loop dynamics (see (58) and (61) respectively) were located at -1.2 and -3 respectively. To this end, the parameters k_{δ_1} , k_{δ_2} , k_{ε_1} and k_{ε_2} were selected as

$$k_{\delta_1} = k_{\delta_2} = 1.2, \quad k_{\varepsilon_1} = 4 \quad \text{and} \quad k_{\varepsilon_2} = 2$$

Fig. 6 shows the evolution of the vehicle's position and orientation during its flight, in the presence of nonconstant wind gusts. The unknown efforts arising from those gusts are estimated via the identifier and are shown in Fig. 7. The magnitude of the aerodynamic efforts F_{ext} and M_{ext} was fixed with the same order as the magnitude of the control signals T_z and τ . In addition, in order to show the performance of the identification structure, the initial conditions of the estimates $\hat{F}_{ext_0}$ and $\hat{M}_{ext_0}$ are different from the true values F_{ext_0} and M_{ext_0} . Fig. 8 shows the behavior of the control inputs. From these figures, it is clear that the designed controller, with the appropriate identification structure, succeeds in reaching the desired position despite arising wind gusts.

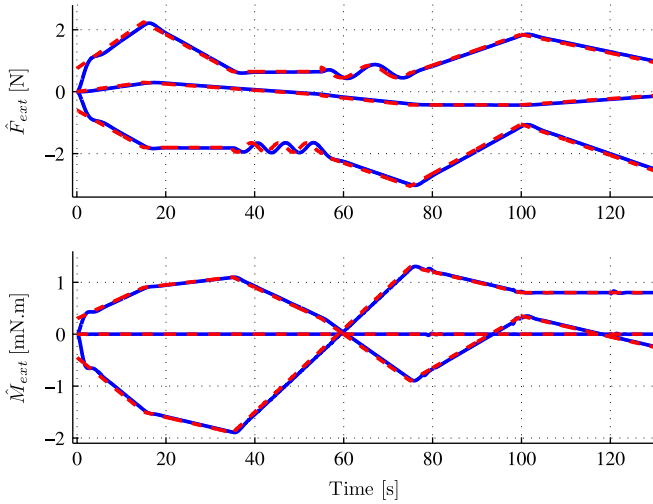


Fig. 7. Estimation of the unknown effort $\hat{F}_{ext}$ and torque $\hat{M}_{ext}$ (solid blue lines) and their true values (dashed red lines). (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)

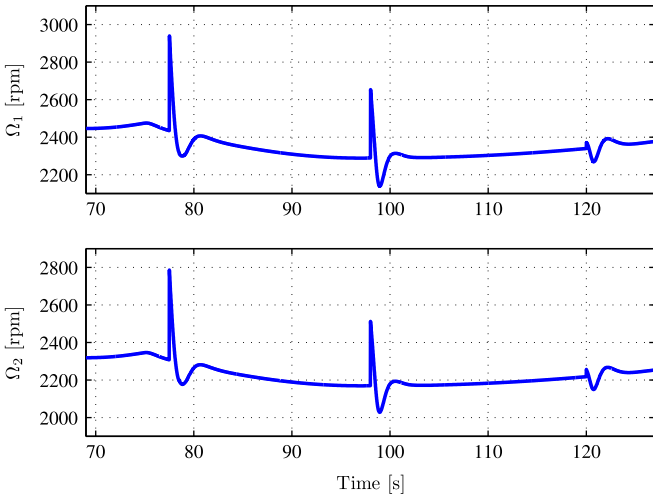


Fig. 8. Control signals Ω_1 and Ω_2 .

The robustness of the hierarchical controller was then tested by considering sensing noise and by varying the system parameters from their nominal value. The behavior of the GLMAV was then investigated in the case of realistic uncertain signals and in the one where the aerodynamic parameters α , β , γ_1 and γ_2 are slightly different from the identified ones. Such a situation can occur, for example, with small mechanical displacements of defects after the gun shot at the beginning of the first flight phase. The lower and upper bounds of uncertainty, noise or delay that can be added to the initial parameters or signals without degrading the control performance too much are listed below:

- for an uncertainty ΔJ on the inertia matrix J : $[-50, 100]$ % of J ;
- for an uncertainty $\Delta \Xi$ on the aerodynamic parameters $\Xi = [\alpha, \beta, \gamma_1, \gamma_2]$: $[-30, 30]$ % of Ξ ;
- for a power spectral density on sensing noise: $[0, 0.005]$;
- for a time delay on the sensors signals: $[0, 50]$ ms;
- for a time delay on the actuators signals: $[0, 150]$ ms;
- for a sampling period: $[0, 300]$ ms.

Depending on the dynamics of the engines, the erratic property of the control signals issued from these simulations may be of

paramount importance while moving from simulations towards the complex real world scenario of the future experimental validation.

6. Experimental results

In this section, the measurements of a corresponding flight experiment on the real system presented in Fig. 2 are presented. For this study, an indoor experimental facility which mitigates many of the complications of sensing, computation and repeatability was designed at the University of Technology of Compiègne. However, before going through the results, the conditions under which the experiments were conducted are introduced.

6.1. Experimental setup

As the flight has been done indoor, no external effects can affect the GLMAV and therefore the identification part of the controller has been disabled. For the same reason, the GPS signals are not available meaning that the position ξ and velocity v measurements must be obtained by a way other than the IMU. The experimental facility is therefore equipped with an OptiTrack motion capture system consisting of 12 cameras, which use infrared light to track reflective markers on the vehicle. One of these cameras is shown in Fig. 9. This system provides robust, reliable, real-time position and velocity tracking of the vehicle with a higher accuracy than would do an IMU. In other words, the position ξ and the velocity v were given by the motion capture system whereas the orientation η and the angular velocity Ω were given by the embedded IMU. The data rate of the IMU being of 100 Hz, the rate for the controller was chosen to be 100 Hz too.

To protect the vehicle from rolling over, flexible arches were placed all around the GLMAV increasing its mass of a hundred grams. In addition, a set of markers covered with a special ultra-reflective coating were suitably attached on the body of the vehicle. The GLMAV is initially placed at the center of the facility allowing the OptiTrack system to lock onto the vehicle at the very beginning of the experiment. The motion capture system then begins relaying position and velocity information to the ground



Fig. 9. An infrared camera of the OptiTrack motion capture system.

station which sends it to the GLMAV by WiFi where the control signal is computed. It provides incredibly clean data for the control algorithm as it is reported in the following section.

6.2. Achievement of the experiments

As in the simulation, the objective of the experiments was to evaluate the ability of the hierarchical control to ensure the proper execution of a position tracking maneuver while maintaining a constant orientation around the yaw angle. The desired trajectory generated by the remote is made up of steps, but only in the horizontal plane. Indeed, the mass of the GLMAV having increased with the protections, the engines are not powerful enough to allow any large movements in the vertical plane any more. In addition, the value of the control parameters k_{δ_1} , k_{δ_2} , k_{ε_1} and k_{ε_2} has been heavily modified to better fit to reality. Indeed, it was found out that the control gains related to the drone orientation should be substantial in order the actuators to be effective. Furthermore, some of these parameters are not simple scalar any more, but 3×3 diagonal matrices allowing to weight each axis independently. These control parameters are now given by

$$k_{\delta_1} = 1.2, \quad k_{\delta_2} = 1.2, \quad k_{\varepsilon_1} = \begin{bmatrix} 45 & 0 & 0 \\ 0 & 45 & 0 \\ 0 & 0 & 10 \end{bmatrix} \quad \text{and} \quad k_{\varepsilon_2} = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 10 \end{bmatrix}. \quad (65)$$

The time response of the closed-loop system with respect to the desired trajectory is plotted in Fig. 10 for the output signals and in Fig. 11 for the control signals.

As expected, the real system shows a slightly worse performance certainly due to small model deviations, particularly in terms of aerodynamics, and noisy data. The loss of efficiency is especially visible on the control of the yaw angle which is not as tight as predicted in the simulations. A reason might be the strong drift of the yaw measurement from the IMU. On the other hand, the horizontal control shows a good correlation with the prediction of the simulation, having only small delay. This can be explained from the latency of the motion capture system and from

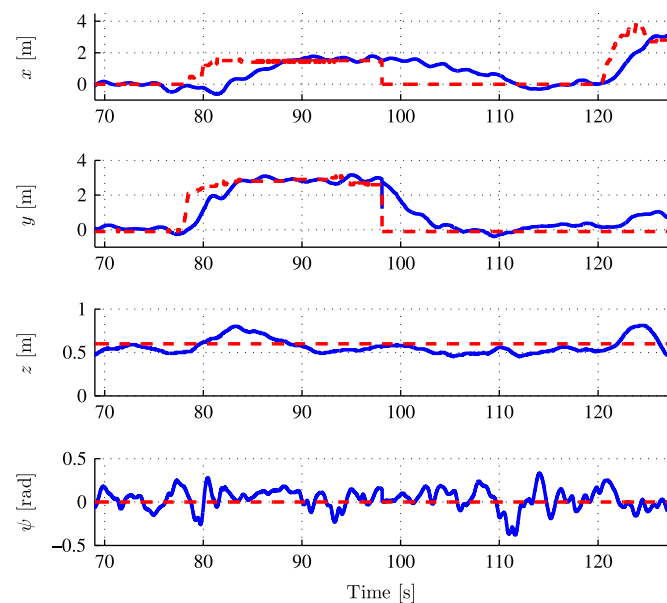


Fig. 10. Output signals x , y , z and ψ (solid blue lines) and reference signals (dashed red lines). (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)

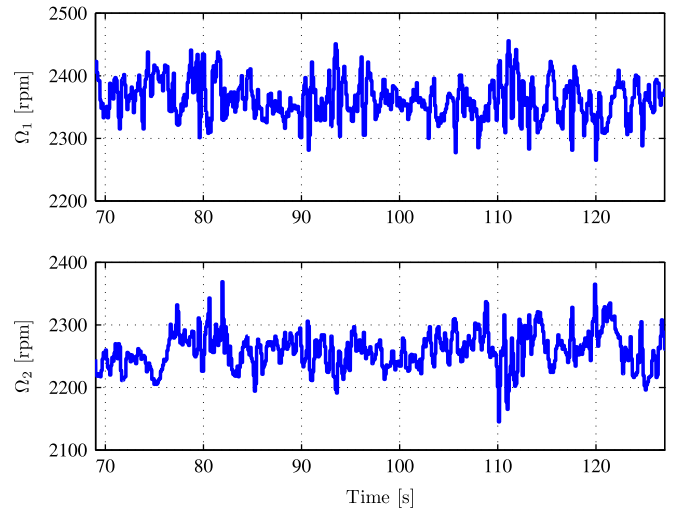


Fig. 11. Control signals Ω_1 and Ω_2 .

the fact that the sensors and actuators have not been modeled. Overall the performance of the control algorithm is very satisfying and encouraging. However, it has to be put into perspective as the experiments were conducted in a friendly environment with the position and the velocity given by a motion capture system. Eventually, only the signals generated by the IMU will have to be used, leading to severe complications in tracking a desired position due to the accuracy of the GPS receiver.

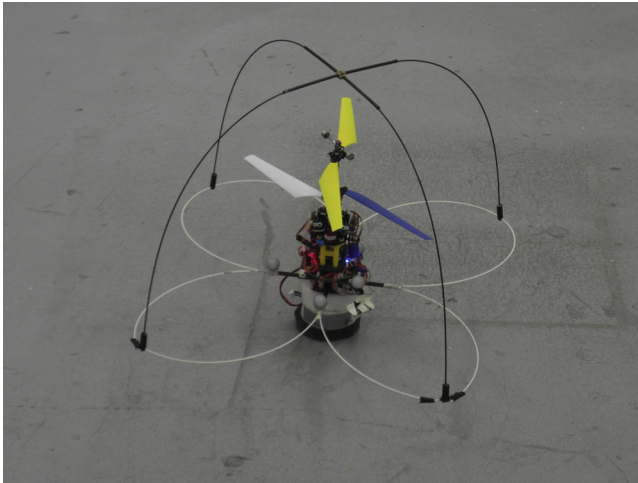
7. Conclusion

In this paper, the project of a Gun Launched MAV investigated at the CRAN, the ISL and the HEUDIASYC is presented. As a first step, the nonlinear model of the dynamics of the vehicle, suitable for control purposes, is developed. That model is based on the rigid body dynamics, where all the possible acting forces and moments are discussed. The paper then focuses on the design of a nonlinear backstepping-based hierarchical controller that includes aerodynamic effects by adding an identification structure. These unknown aerodynamic effects are assumed to be constant (or slowly time-varying). However, it is shown by the numerical simulations that the identifier can handle aerodynamic effects that vary to some extent. The controller is finally successfully tested on the real system in a controlled environment, showing the appropriateness of the overall design procedure.

As future work, the performance of the control law might be improved by accurately tuning the control gains. In addition, there is the idea to model the sensors and the actuators in order to try to get rid of the delay in the time responses. The last task is to perform an outdoor flight, provided that the position measurement is available and accurate enough. Of course, several control approaches will be experimentally tested (PID, $\mathcal{H}_\infty$, etc.) in order to pick the one which exhibits the best tracking performance.

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Video S1. Indoor flight experiment for the GLMAV.

A video clip is available online. Supplementary material related to this article can be found online at <http://dx.doi.org/10.1016/j.conengprac.2013.11.016>

Appendix A. Supplementary material

The following are the supplementary data to this article:
[Video S1.](#)

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